

Target-Local Fusion for Communication-Efficient Multi-UAV OTFS-ISAC Sensing

Xinhang Tan, Zhengquan Zhang, L Hao
Southwest Jiaotong University
Chengdu, China

Email: tanxinhang6478@163.com, zhang.zhengquan@swjtu.edu.cn, lhao@home.swjtu.edu.cn

Abstract—Communication-constrained cooperative sensing depends on both the selected observations and the locations where their soft information is fused. We propose target-local fusion for prior-assisted multi-UAV OTFS-ISAC sensing. Each target is assigned to the UAV nearest its predicted position, after which a detector-aligned adaptive coarse-to-fine (C2F) selector chooses local observations and finite-blocklength remote reports. In 1000 paired trials with 15 UAVs and 10 targets, the proposed method achieved $P_D = 0.9764$ and a worst-target P_D of 0.966. Its detection difference from sensing-SINR selection was not statistically significant, while it reduced the mean payload and serial reporting delay by 83.7%. A fusion-rule ablation showed that this saving primarily arose because target-local planning retained more evidence at its fusion UAV. The result is limited to prior-assisted sensing with the stated finite-blocklength and serial-reporting model; performance decreased under large prediction errors and sparse deployments.

Index Terms—Multi-UAV networks, OTFS-ISAC, target-local fusion, cooperative sensing, finite-blocklength reporting, coarse-to-fine selection.

I. INTRODUCTION

Integrated sensing and communication (ISAC) reuses spectrum and hardware for communication and sensing [1], [2]. UAV mobility extends coverage [5], [6], while cooperation supplies complementary target observations [3].

Cooperation nevertheless creates a reporting problem. An observation follows the bistatic sensing chain $i \rightarrow q \rightarrow j$, but its soft statistic must subsequently reach a fusion UAV. Its value therefore depends on both sensing quality and reporting reliability [9]–[11]. Existing node-selection and backhaul-aware methods address parts of this coupling, while distributed detection models often abstract the OTFS delay–Doppler (DD) geometry [4], [7], [8].

Backhaul-aware selection optimizes reporting to a fusion center [9]; target-handover methods exchange track information across base stations [14], [15]. We instead study per-target fusion placement before bistatic OTFS observation selection. The resulting local/remote classification changes packet count, delivery, and detector-aligned C2F. The contribution is this coupling and its controlled evaluation.

We study prior-assisted confirmation or re-detection and introduce target-local fusion. For each target, the method

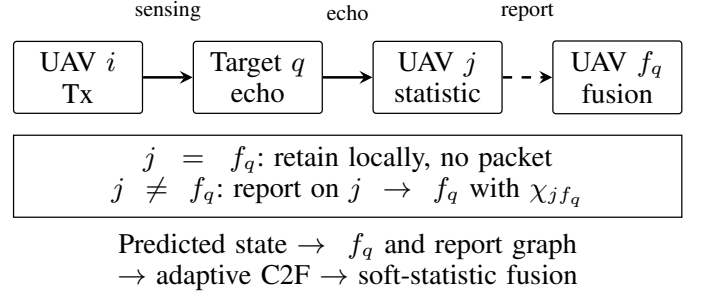


Fig. 1. Sensing and reporting have different endpoints. The fusion UAV is chosen from predicted target state before observation selection and reporting.

chooses the UAV nearest its predicted position, without accessing the current target truth. It then constructs the target-specific reporting graph and applies a detector-aligned adaptive C2F selector. This decomposition separates fusion placement from observation selection. The outer distance rule is a reproducible heuristic rather than a joint optimum.

Our contributions are threefold.

- We introduce prediction-only target-local fusion for multi-UAV bistatic OTFS-ISAC. The target-specific fusion UAV converts its own observations into local evidence and exposes an explicit remote-reporting graph.
- We integrate this architecture with a detector-aligned adaptive C2F selector. The selector accounts for finite-blocklength delivery and reduces the number of waveform-level DD refinements relative to full refinement.
- We isolate the architectural effect through a 1000-trial paired comparison, a controlled fusion-rule ablation, and operating-boundary sweeps. The evidence supports communication savings with no statistically resolved detection difference from Sensing-SINR in the tested setting.

II. SYSTEM MODEL

A. Task, Prior Information, and Reporting Graph

Let $\mathcal{M} = \{1, \dots, M\}$ and $\mathcal{Q} = \{1, \dots, Q\}$ denote UAVs and targets. Observation $e = (i, j, q)$ is generated at echo receiver j :

$$i \rightarrow q \rightarrow j \rightarrow f_q. \quad (1)$$

The sensing pair is (i, j) , whereas its reporting link is (j, f_q) . All statistics for target q meet at one fusion UAV f_q . When $j = f_q$, no report is transmitted (Fig. 1).

The task is confirmation or re-detection after upstream acquisition. An external tracker supplies $b_q = \mathcal{N}(\hat{\mathbf{x}}_q, \mathbf{P}_q)$; we simulate its output with noisy horizontal position and velocity estimates, keeping nominal target altitude known. Only this belief and the mean RCS enter planning. The unknown current state enters echo generation and final detection. The DD search gate propagates $\mathbf{P}_q$ through the bistatic Jacobian and uses three standard deviations per DD coordinate. This gate models search coverage, not a posterior tracker or blind initial acquisition.

B. Reporting Reliability and Cost

Sensing and reporting share power, $P_i^s = \rho P_i$ and $P_i^c = (1 - \rho)P_i$. The UAV channel gain is $g_{ab}^c = (\lambda/4\pi)^2 d_{ab}^{-\alpha_c} \zeta_{ab} |h_{ab}^c|^2$, with shadowing ζ_{ab} and fading h_{ab}^c . Under orthogonal serial reporting,

$$\gamma_{ab}^c = \frac{P_a^c g_{ab}^c}{N_0 + \epsilon_s \sum_{k \neq a, b} P_k^s g_{kb}^c + \epsilon_d P_a^s g_{ab}^c}, \quad (2)$$

$$R_{ab} = B_c \log_2(1 + \gamma_{ab}^c).$$

Here N_0 is receiver noise power over B_c , and ϵ_s, ϵ_d are residual sensing leakage factors. Reports do not mutually interfere; sensing transmissions remain concurrent.

For a k -bit packet in n channel uses, we adopt the normal approximation for a complex AWGN-equivalent reporting channel [12]:

$$\chi_{ab} \simeq 1 - \mathcal{Q}\left(\frac{C(\gamma_{ab}^c) - k/n}{\sqrt{V(\gamma_{ab}^c)/n}}\right), \quad (3)$$

where $C(\gamma) = \log_2(1 + \gamma)$ and $V(\gamma) = [1 - (1 + \gamma)^{-2}](\log_2 e)^2$. This conditional packet-success approximation replaces the heuristic $\gamma/(\gamma + \gamma_{\text{req}})$; it is not a measured code-specific error curve. A remote link must satisfy the range mask, $R_{ab} \geq R_{\min}$, and $\chi_{ab} \geq \chi_{\min}$. Each selected remote observation reserves one fixed padded k -bit report and n/B_c seconds, without aggregation or retransmission. Thus N_{rem} reports cost kN_{rem} bits and nN_{rem}/B_c serial delay. Local delivery has $\chi = 1$ and zero reporting cost; sensing, processing, and unmodeled control traffic are not free.

C. DD Energy and Local Statistic

With $\mathbf{u}_{iq}$ pointing from UAV i to target q , the bistatic coordinates are

$$\begin{aligned} \tau_{ijq} &= (d_{iq} + d_{jq})/c, \\ \nu_{ijq} &= \lambda^{-1}[(\mathbf{v}_q - \mathbf{v}_i)^T \mathbf{u}_{iq} + (\mathbf{v}_q - \mathbf{v}_j)^T \mathbf{u}_{jq}]. \end{aligned} \quad (4)$$

Let $\ell = \tau L \Delta f$ and $\kappa = \nu NT$, with $T = 1/\Delta f$. The coarse estimate is $\eta^c = \text{sinc}^2(\ell - \text{round}(\ell))\text{sinc}^2(\kappa - \text{round}(\kappa))$. For a periodic A -bin axis define

$$D_A(x) = \frac{\sin(\pi x)}{A \sin(\pi x/A)}, \quad p_A(b; z) = \frac{|D_A(b - z)|^2}{\sum_{a=0}^{A-1} |D_A(a - z)|^2},$$

$$\eta_{ijq}^{\text{loc}} = \sum_{(b,d) \in \mathcal{W}_W(\kappa, \ell)} p_N(b; \kappa) p_L(d; \ell). \quad (5)$$

The window contains $(2W + 1)^2$ bins around the rounded center, with indices wrapped modulo (N, L) . The removable singularity of D_A is evaluated by continuity. This normalized finite-grid leakage model gives $0 \leq \eta^{\text{loc}} \leq 1$. We use $W = 1$ (nine bins) and $\eta^f = \eta^{\text{loc}}$, with no interpolation coefficient or gain floor. The fractional-center cache resolution is 10^{-3} bin. The compact OTFS chain [13] is checked separately below; $W = 1$ is a fixed processing choice, not a proved optimal window.

With mean target RCS $\bar{\sigma}_q$, $g_{ijq}^s = \lambda^2 \bar{\sigma}_q / [(4\pi)^3 d_{iq}^2 d_{jq}^2]$, and

$$\gamma_{ijq}^s = \frac{P_i^s g_{ijq}^s G_{ij}^{\text{hw}} G_p \eta_{ijq}^{\text{DD}} \eta_{ijq}^{\text{col}}}{N_0 + I_j^s}. \quad (6)$$

Here G_{ij}^{hw} contains antenna gains and losses, and I_j^s includes residual self-interference and concurrent direct paths. The collision surrogate is $\eta^{\text{col}} = 1/n_{ijq}$, where n_{ijq} counts targets in the same rounded DD bin for pair (i, j) . It is an occupancy penalty, not a coherent multi-target interference law.

For K_{look} independent looks, normalized energy X has Gamma shape K_{look} and scale 1 under $\mathcal{H}_0$, or $1 + \gamma$ under $\mathcal{H}_1$. The centered LLR is

$$\ell_e = \frac{\gamma}{1 + \gamma} (X - K_{\text{look}}). \quad (7)$$

Consequently, $\delta_e = K_{\text{look}} \gamma^2 / (1 + \gamma)$, $v_{e,0} = K_{\text{look}} \gamma^2 / (1 + \gamma)^2$, and $v_{e,1} = K_{\text{look}} \gamma^2$. Look-to-look fluctuation is marginalized by this energy law rather than sampled again in the mean-RCS link budget.

For successful delivery $A_e \sim \text{Bernoulli}(\chi_e)$, set $Y_e = A_e \ell_e + (1 - A_e) Z_e$, with independent $Z_e \sim \mathcal{N}(0, av_{e,0})$. Total expectation and variance give

$$\begin{aligned} \delta_e^{\text{eff}} &= \chi_e \delta_e, \\ \bar{v}_{e,0} &= [\chi_e + (1 - \chi_e) a] v_{e,0}, \\ \bar{v}_{e,1} &= \chi_e v_{e,1} + (1 - \chi_e) a v_{e,0} + \chi_e (1 - \chi_e) \delta_e^2. \end{aligned} \quad (8)$$

The archived V1 implementation uses $a = 9$: failed reports are replaced by zero-mean uncertainty. This Gaussian replacement is an explicit surrogate, not a consequence of finite blocklength or a true erasure ($a = 0$). The between-component term in $\bar{v}_{e,1}$ cannot be omitted.

For independent selected observations, use $w_e \propto \delta_e^{\text{eff}} / \bar{v}_{e,0}$, $m_{q,1} = \sum_{e \in \mathcal{S}_q} w_e \delta_e^{\text{eff}}$, and $v_{q,h} = \sum_{e \in \mathcal{S}_q} w_e^2 \bar{v}_{e,h}$. The selector predicts

$$t_q = \sqrt{v_{q,0}} \left[z_0 + \frac{\kappa_{q,0}}{6} (z_0^2 - 1) \right],$$

$$\hat{P}_{D,q} = \mathcal{Q}\left(\frac{t_q - m_{q,1}}{\sqrt{v_{q,1}}}\right), \quad (9)$$

where $z_0 = Q^{-1}(P_{FA})$ and $\kappa_{q,0}$ is null skewness. Empty-set prediction is zero. Scheduling uses belief-based moments; final Monte Carlo detection uses truth-side observations and the same threshold rule. This remains a moment approximation, not an exact detection probability. The supplement documents the archived failure-model naming and validation.

III. TARGET-LOCAL ADAPTIVE C2F SELECTION

A. Prediction-Only Fusion Placement

The outer stage assigns each target to the UAV nearest its predicted position:

$$f_q = \arg \min_{m \in \mathcal{M}} \|\hat{\mathbf{p}}_q - \mathbf{p}_m^{\text{UAV}}\|_2. \quad (10)$$

The rule uses neither the current target truth nor its realized RCS. It is a low-complexity placement heuristic, not a jointly optimal fusion solution. Given f_q , the feasible observation set is

$$\mathcal{E}_q = \{(i, j, q) : (i, j, q) \text{ is DD-valid, } j = f_q \text{ or } (j, f_q) \text{ is feasible}\}. \quad (11)$$

Local-delivery bound. Under (8), the single-observation deflection is $d(\chi) = \chi^2 \delta^2 / [v_0(a + (1-a)\chi)]$. Its derivative is nonnegative for $a \geq 0$, so local delivery maximizes this quantity for the same observation. This does not imply monotonicity of the approximate P_D or optimality of nearest-target placement when other report links change.

For fixed observation laws and hypothesis-independent replacement, lower reliability is a random degradation of higher reliability. Optimal detection at fixed false alarm is therefore nondecreasing with reliability [16]. The supplementary proof concerns the optimal ROC envelope, not the implemented linear detector.

For a fixed S_q , let a_{qm} count selected observations received at UAV m . Every fusion choice obeys $N_{\text{rem},q} \geq |S_q| - \max_m a_{qm}$. This follows because only observations produced at the fusion UAV avoid a packet. A feasible nearest-target assignment attaining the bound is report-minimal for that fixed set; no joint selection optimum is implied.

B. Communication-Constrained Selection

Let $\hat{\mathbf{P}}_D(S)$ denote the detector-predicted target probabilities from (9). We cap each target probability at the design requirement and define a smooth weak-target utility

$$U(\hat{\mathbf{P}}_D) = -Q\tau_\alpha \log \sum_q \exp[-\bar{P}_{D,q}/\tau_\alpha] - \frac{\mu_\alpha}{2P_D^{\text{req}}} \sum_q [P_D^{\text{req}} - \hat{P}_{D,q}]_+^2, \quad (12)$$

where $\bar{P}_{D,q} = \min(\hat{P}_{D,q}, P_D^{\text{req}})$. The first term prioritizes weak targets, while the second penalizes unresolved detection gaps. The selector heuristically seeks a high value of the capped objective:

$$\begin{aligned} \max_{S \subseteq \cup_q \mathcal{E}_q} F(S) &= U(\hat{\mathbf{P}}_D(S)) - \lambda_c \sum_{e \in S} c_e, \\ \text{s.t. } |S_q| &\leq L_{\max}, \quad |S| \leq L_{\text{tot}}, \end{aligned} \quad (13)$$

TABLE I
PREDICTION-ONLY ADAPTIVE C2F (RELEASED V1)

Require: Belief-side tables, f_q , caps, $J = 25$, $W = 1$

- 1: Build triplet sets $\mathcal{E}_q$; set $S = \emptyset$, $H_q = \emptyset$.
- 2: **while** $|S| < L_{\text{tot}}$ **do**
- 3: Compute coarse $\Delta F(e | S)$ for unselected candidates satisfying caps and $\Delta D_q > 0$.
- 4: For each target, add its highest positive-gain candidate outside H_q if $|H_q| < \min(J, |\mathcal{E}_q|)$.
- 5: Let e^* have largest positive gain; stop if none exists.
- 6: Commit e^* to S . Insert it into H_q if absent, replacing the last entry when full.
- 7: **end while**
- 8: Evaluate η^{loc} only on $H = \cup_q H_q$.
- 9: Reset $S = \emptyset$; greedily add the largest positive fine ΔF on H , checking caps and $\Delta D_q > 0$ each time.
- 10: **return** S when no admissible positive gain remains or the total cap binds.

where $c_e = 10^3 \mathbf{1}\{j \neq f_q\} n / B_c$ is measured in milliseconds, matching the implemented price λ_c . Fixed blocklength makes c_e a remote-report cardinality price rather than a link-dependent delay. The caps limit observations, including local ones; the price does not enforce a prescribed hard communication budget. They imply payload at most kL_{tot} and serial delay at most nL_{tot}/B_c . The greedy replay repeatedly commits the feasible candidate with the largest positive marginal $F(S \cup \{e\}) - F(S)$. Because the objective can be non-monotone and the constraints combine target and total caps, we claim neither global optimality nor a universal submodular approximation ratio.

C. Adaptive C2F Acceleration

The pseudocode in Table I constructs a per-target frontier, then replays the same objective on its fine estimates. The per-target shortlist cap is $J = 25$ and the observation cap is six. A positive-deflection filter uses $D_q = \sum_{e \in S_q} d(\chi_e)$ under independence. Cached moments are invalidated when their target's selected set changes. Ties retain the first candidate in target/transmitter/receiver order.

Let E be the number of feasible candidates and r the number refined. If a coarse evaluation costs C_c and a fine evaluation costs C_f , full refinement costs EC_f , whereas adaptive C2F has DD evaluation cost approximately

$$EC_c + rC_f, \quad r \leq E. \quad (14)$$

This excludes greedy scoring and frontier maintenance. With at most L_{tot} commits per stage, scanning requires $O(L_{\text{tot}}(E + r))$ candidate-score evaluations; a direct implementation costs $O(L_{\text{tot}}(E + r)(L_{\max} + Q))$ for moments and global utility evaluation, before caching. Nearest placement costs $O(QM)$ and the shortlist stores at most QJ triplets. When fine DD refinement dominates, the fractional reduction approaches $1 - r/E$. The main experiment used 61.3 refinements for 860.2 feasible candidates (92.9% fewer). This is a work-count ratio, not a wall-clock speedup. No bound on omitted candidates is available, so C2F is not claimed lossless.

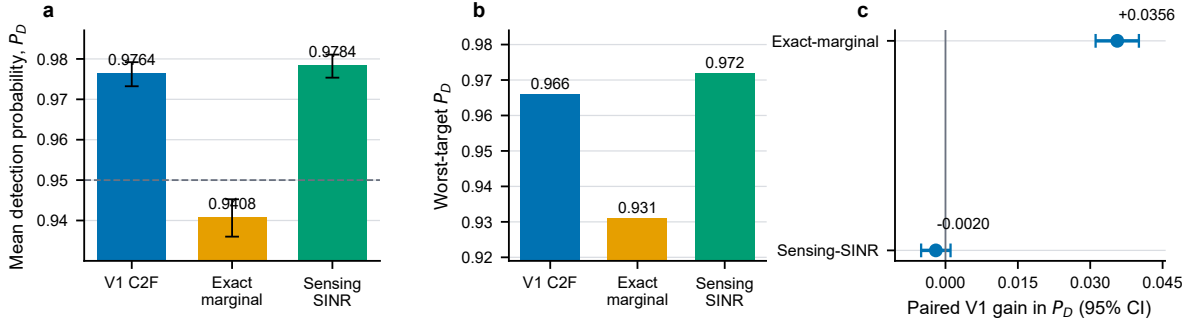


Fig. 2. Confirmatory MC=1000 comparison under a common scenario and caps. (a) Mean detection probability with 95% Wilson intervals; the dashed line marks 0.95. (b) Worst-target detection probability. (c) Paired V1-minus- baseline differences with 95% confidence intervals.

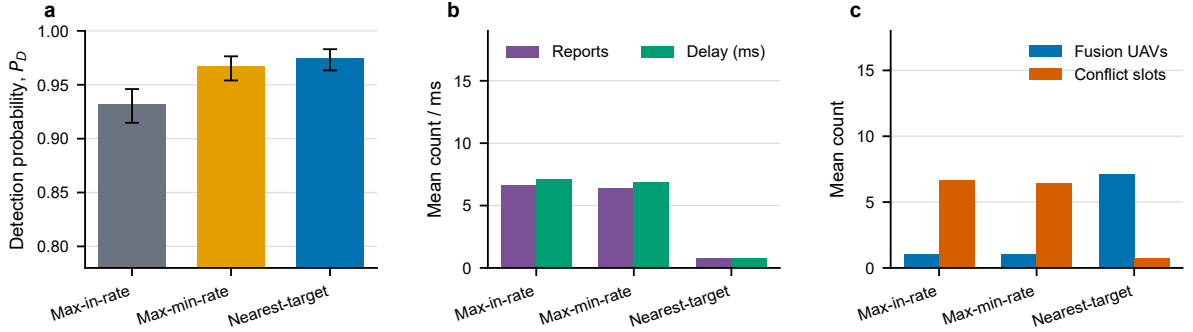


Fig. 3. Fusion-rule mechanism ablation (MC=100 per rule) with the inner adaptive C2F selector fixed. (a) Detection probability and 95% intervals. (b) Selected reports and modeled serial delay. (c) Number of assigned fusion UAVs and conflict-graph slots. The slot count diagnoses parallelization potential; it is not an achieved latency.

IV. SIMULATION RESULTS

A. Protocol and Nominal Configuration

We evaluate prior-assisted confirmation in a $4000\text{ m} \times 4000\text{ m}$ region with $M = 15$ UAVs and $Q = 10$ targets. Table II lists the released V1 configuration. All main methods use the same paired geometries, detector randomness, physical model, and observation caps. The confirmatory comparison uses 1000 Monte-Carlo trials. Fusion and operating-envelope sweeps use 100 trials per condition with seed 2026 and are therefore diagnostic rather than substitutes for the main evidence.

B. Detection Performance and Communication Cost

Figure 2 compares target-local adaptive C2F with two strong rules. Exact-marginal greedy maximizes each step's deflection-based utility gain without a delay price, with total observations capped at V1's count and six per target. "Exact" refers to marginal evaluation, not global optimality. Sensing-SINR ranks coarse sensing SINR and matches V1's per-target observation counts. Both use V1's fusion graph and feasibility checks; remote-report counts are intentionally outcomes, not matched constraints. V1 attains $P_D = 0.9764$ (95% CI: $[0.9732, 0.9792]$), versus 0.9408 for exact-marginal greedy and 0.9784 for Sensing-SINR. Worst-target values are 0.966,

TABLE II
TARGET-LOCAL FUSION V1 SIMULATION PARAMETERS

Parameter	Value
UAVs / targets	$M = 15, Q = 10$
Area / communication range	$4000^2 \text{ m}^2, d_{\max} = 2500 \text{ m}$
UAV / target speed	$30\text{--}60 / 50\text{--}90 \text{ m/s}$
Carrier / DD grid	$5.9 \text{ GHz}, N = L = 64, \Delta f = 30 \text{ kHz}$
Power split / transmit power	$\rho = 0.8, P_i = 1 \text{ W}$
Rate threshold / false alarm	$R_{\min} = 0.2 \text{ Mbit/s}, P_{\text{FA}} = 0.05$
Report / blocklength	$640 \text{ bits}, n = 2048 \text{ channel uses}$
Predicted belief	$\sigma_p = 150 \text{ m}, \sigma_v = 15 \text{ m/s}$
Selection	$P_D^{\text{req}} = 0.95, \lambda_c = 0.005, L_{\max} = 6, L_{\text{tot}} = 60$
Fusion / reporting	nearest predicted target / serial orthogonal MAC
RCS / looks / failure model	$50 \text{ m}^2 / 16 / \text{Gaussian replacement}, a = 9$
C2F / utility	$W = 1, J = 25, \tau_\alpha = 0.1, \mu_\alpha = 1$
Main / sweep trials	$1000 / 100 \text{ per condition}$

0.931, and 0.972. The paired V1-minus-baseline intervals are $+0.0356$ ($[0.0311, 0.0401]$) and -0.0020 ($[-0.0050, 0.0010]$), respectively. Thus, V1 improved on exact-marginal selection; its detection difference from Sensing-SINR was not statistically resolved. This does not establish equivalence or non-inferiority. The supported headline is communication effi-

ciency with no detected significant change in detection.

V1 selects 12.101 observations and sends 0.751 remote reports on average, corresponding to 0.481 kbit and 0.801 ms. Sensing-SINR selects the same 12.101 observations but sends 4.595 reports, 2.941 kbit, and 4.901 ms. V1 therefore reduces payload and serial delay by 83.7% without a significant P_D change. Both savings follow from the same report-count reduction under fixed packet size and serial MAC. A separate MC=200 full-refinement control uses the same detector-PD greedy rule. Adaptive C2F uses 61.06 versus 853.65 fine evaluations (92.85% fewer), with $P_D = 0.9775$ versus 0.9715. The paired difference is 0.0060 ([0.0005, 0.0115]); full refinement is a computational control, not an oracle.

C. Fusion-Placement Ablation

Figure 3 changes only the fusion destination rule. The target-agnostic max-in-rate rule gives $P_D = 0.932$, worst-target $P_D = 0.900$, 6.65 remote reports, and 7.093 ms. A max-min-rate alternative gives 0.967, 0.940, 6.40, and 6.827 ms. Nearest-target fusion reaches 0.975 and 0.960 with only 0.74 remote reports and 0.789 ms. It also reduces selected observations from 19.89 and 15.61 to 12.24. The nearest-target rule therefore converted more selected evidence into local observations with unit reliability and no packet cost.

Nearest-target fusion distributes targets across 7.13 fusion UAVs on average, versus one for the other rules. Its 0.70 conflict-graph slots indicate possible parallelism, not achieved delay; all reported delays use serial MAC.

D. Operating Boundaries

Increasing position error from 50 to 500 m reduced P_D from 0.988 to 0.906. Increasing deployment side from 2500 to 5500 m reduced it from 0.989 to 0.940 and increased reports from 0.05 to 2.67. Other 100-trial scans in the supplement calibrate parameters without supporting universal recommendations.

E. Component Validation and Runtime

A separate audit compares the normalized DD expression with a compact OTFS impulse-response simulation at 512 fractional offsets. The maximum local-window energy discrepancy is 3.2×10^{-15} ; the coarse sinc approximation differs by at most 1.06×10^{-3} . Six waveform-derived LLR tests with 10^5 draws each give at most 0.00184 discrepancy from exact single-report mixture detection, but up to 0.0447 from the moment-based prediction. This supports the component law while exposing the scheduling approximation's limits, not validating a complete multi-target radio. On 30 additional scenarios, median selector time is 0.556 s for adaptive C2F and 2.032 s for full refinement on a Ryzen 7 7800X3D. Timing excludes shared table construction and precomputed DD gains; it is not end-to-end latency. The supplement supplies seeds, intervals, and limitations.

V. CONCLUSION

Target-local fusion retains evidence at its producing UAV before adaptive C2F selects remote reports. In 1000 paired trials, it achieved $P_D = 0.9764$ with no statistically resolved detection difference from Sensing-SINR and 83.7% less payload and serial reporting delay. The contribution is architectural communication efficiency under independent LLRs, fixed packets, and prior-assisted sensing. Placement remains heuristic, detector prediction approximate, and performance sensitive to belief error and sparse deployment.

REFERENCES

- [1] F. Liu *et al.*, "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, Jun. 2022, doi: 10.1109/JSAC.2022.3156632.
- [2] N. González-Prelcic *et al.*, "The integrated sensing and communication revolution for 6G: Vision, techniques, and applications," *Proc. IEEE*, vol. 112, no. 7, pp. 676–723, Jul. 2024, doi: 10.1109/JPROC.2024.3397609.
- [3] K. Meng, C. Masouros, A. P. Petropulu, and L. Hanzo, "Cooperative ISAC networks: Performance analysis, scaling laws, and optimization," *IEEE Trans. Wireless Commun.*, vol. 24, no. 2, pp. 877–892, Feb. 2025, doi: 10.1109/TWC.2024.3491356.
- [4] K. Meng, C. Masouros, G. Chen, and F. Liu, "Network-level integrated sensing and communication: Interference management and BS coordination using stochastic geometry," *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 19365–19381, Dec. 2024, doi: 10.1109/TWC.2024.3483031.
- [5] J. Mu, R. Zhang, Y. Cui, N. Gao, and X. Jing, "UAV meets integrated sensing and communication: Challenges and future directions," *IEEE Commun. Mag.*, vol. 61, no. 5, pp. 62–67, May 2023, doi: 10.1109/MCOM.008.2200510.
- [6] K. Meng *et al.*, "UAV-enabled integrated sensing and communication: Opportunities and challenges," *IEEE Wireless Commun.*, vol. 31, no. 2, pp. 97–104, Apr. 2024, doi: 10.1109/MWC.131.2200442.
- [7] A. Chawla, A. Patel, A. K. Jagannatham, and P. K. Varshney, "Distributed detection in massive MIMO wireless sensor networks under perfect and imperfect CSI," *IEEE Trans. Signal Process.*, vol. 67, no. 15, pp. 4055–4068, Aug. 2019, doi: 10.1109/TSP.2019.2924588.
- [8] A. Chawla, A. S. Sarode, A. K. Jagannatham, and L. Hanzo, "Distributed parameter detection in massive MIMO wireless sensor networks relying on imperfect CSI," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 506–519, Jan. 2021, doi: 10.1109/TWC.2020.3025877.
- [9] M. Chen, M.-M. Zhao, A. Liu, M. Li, and Q. Shi, "Joint node selection and resource allocation optimization for cooperative sensing with a shared wireless backhaul," *IEEE Trans. Signal Process.*, vol. 73, pp. 67–82, 2025, doi: 10.1109/TSP.2024.3516709.
- [10] W. Yuan *et al.*, "From OTFS to DD-ISAC: Integrating sensing and communications in the delay-Doppler domain," *IEEE Wireless Commun.*, vol. 31, no. 6, pp. 152–160, Dec. 2024, doi: 10.1109/MWC.018.2300607.
- [11] M. F. Keskin *et al.*, "Integrated sensing and communications with MIMO-OTFS: ISI/ICI exploitation and delay-Doppler multiplexing," *IEEE Trans. Wireless Commun.*, vol. 23, no. 8, pp. 10229–10246, Aug. 2024, doi: 10.1109/TWC.2024.3370501.
- [12] Y. Polyanskiy, H. V. Poor, and S. Verdú, "Channel coding rate in the finite blocklength regime," *IEEE Trans. Inf. Theory*, vol. 56, no. 5, pp. 2307–2359, May 2010, doi: 10.1109/TIT.2010.2043769.
- [13] R. Hadani *et al.*, "Orthogonal time frequency space modulation," in *Proc. IEEE WCNC*, 2017, pp. 1–6, doi: 10.1109/WCNC.2017.7925924.
- [14] Y. Ge *et al.*, *Target handover in distributed integrated sensing and communication*, arXiv:2411.01871, 2024.
- [15] L. Bai, Y. Ge, and H. Wymeersch, *Belief propagation-based target handover in distributed integrated sensing and communication*, arXiv:2506.23118, 2025.
- [16] D. Blackwell, *Equivalent comparisons of experiments*, *Ann. Math. Stat.*, vol. 24, no. 2, pp. 265–272, 1953, doi: 10.1214/aoms/117729032.